

29/05/10909

Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech

Code:100304

Subject: Data Structure & Algorithms

Time: 03 Hours

Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer from the following (Any seven question only):-

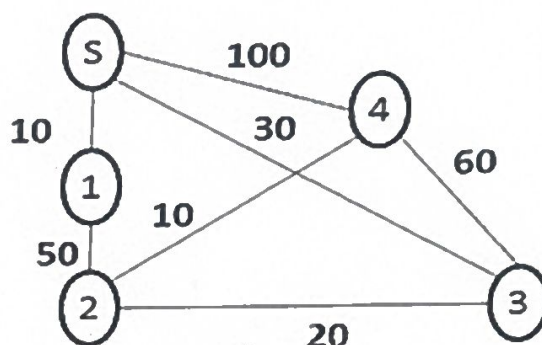
[2 x 7 = 14]

- (a) Which of the following is not a linear data structure?
 - (i) Stack
 - (ii) Array
 - (iii) Queue
 - (iv) Tree
- (b) Which asymptotic notation gives the best case complexity?
 - (i) Big Theta
 - (ii) Big Omega
 - (iii) Big O
 - (iv) Little o
- (c) In stack, which operation is not allowed directly?
 - (i) Push
 - (ii) Peek
 - (iii) Pop
 - (iv) Deletion from bottom
- (d) Which data structure is used in recursive function calls?
 - (i) Queue
 - (ii) Linked List
 - (iii) Stack
 - (iv) Array
- (e) Which queue allows insertion and deletion at both ends?
 - (i) Circular queue
 - (ii) Deque
 - (iii) Simple queue
 - (iv) Priority queue
- (f) Which of the following expression is in postfix form?
 - (i) $A + B$
 - (ii) $A \cdot B \cdot C * +$
 - (iii) $A + B * C$
 - (iv) $(A + B) * C$
- (g) Which of the following has both forward and backward traversal?
 - (i) Singly Linked List
 - (ii) Circular Queue
 - (iii) Doubly Linked List
 - (iv) Simple Queue
- (h) Which of the following trees is height-balanced?
 - (i) Binary Tree
 - (ii) AVL Tree
 - (iii) B-Tree
 - (iv) Threaded Tree
- (i) What is the time complexity for insertion at the beginning of a singly linked list?
 - (i) $O(1)$
 - (ii) $O(n)$
 - (iii) $O(\log n)$
 - (iv) $O(n \log n)$
- (j) Which of the following uses divide and conquer strategy?
 - (i) Insertion Sort
 - (ii) Merge Sort
 - (iii) Bubble Sort
 - (iv) Linear Search

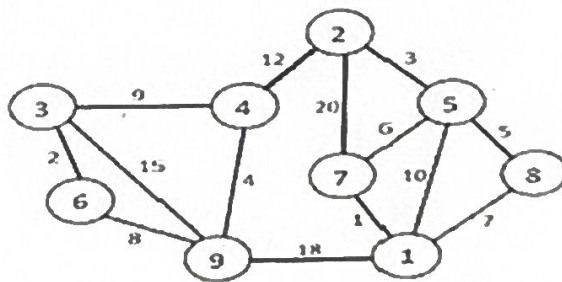
- Q.2 (a) Assume the declaration of multi-dimensional arrays A and B to be, A (-2:2, 2:22) and B (1:8, -5:5, -10:5) [7]
- (i) Find the length of each dimension and number of elements in A and B.
- (ii) Find the address of element B (2, 2, 3), assuming Base address of B =400 and there are W=4 words per memory location. [7]
- (b) What is Sparse matrix? Explain how a Sparse matrix can be implemented by using the linked list? [7]

- Q.3 (a) Write an algorithm for Insertion Sort. Use Insertion sort algorithm, sort the following elements: [7]
- 2, 8, 7, 1, 3, 5, 6, 4.
- (b) What is circular linked list? Write an algorithm to delete a node from begin in single linked list. [7]

- Q.4 (a) Consider the following infix expression and convert it into postfix using stack [7]
- $A + (B * C - (D/E - F) * G) * H$
- (b) Find the shortest path from S to all remaining vertices of graph using Dijkstra Algorithm [7]



- Q.5 (a) Define Hashing. Explain the mid-square and digit folding method hash function with the help of an example. [7]
- (b) Use Heap sort algorithm to sort the following sequence {8, 5, 45, 24, 36, 11, 43, and 21}. [7]
- Q.6 (a) Discuss spanning tree. Write down the Kruskal algorithm to obtain a minimum cost spanning tree. Use Kruskal algorithm to find the minimum cost spanning tree in the following graph: [7]



- (b) What do you mean by priority queue? Explain the types to maintain the priority queue in memory? [7]

- Q.7 (a) Explain B-tree. Write down the properties of it. Consider following sequence of inputs: [7]
- 10,20,35,40,50,60,75,80,95
- Assume that the order of the B-tree is 3.
- (b) Draw a binary tree with following traversals: [7]
- Preorder: A B C D E F G H I J K L
- Postorder: C F E G D B K J L I H A

- Q.8** (a) Do the following operations for constructing a BST [7]
(i) 45, 37, 98, 76, 13, 39, 105, 80, 5 insert element as per their occurrence.
(ii) Delete 39 and 45 respectively
Now Traverse the final BST in Inorder, Preorder and Postorder.
- (b) Describe an AVL tree. Construct an AVL tree by inserting the following elements [7]
in the order of their occurrence {60, 2, 15, 20, 12, 115, 90 and 88}.
- Q.9** (a) Write merge sort algorithm and its analysis. Use merge sort algorithm to sort [7]
9, 11, 10, 1, 60, 10, 6, 25, 40, and 30. Is it a stable sorting algorithm? Justify.
- (b) Write short notes on *any one* the following: [7]
(i) Double-ended Queue (Deque)
(ii) Threaded Binary Tree
(iii) Time Complexity and Big-O Notation
(iv) DFS, BFS



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